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B.Tech. Degree VI Semester Examination April 2018

ME 606 CAD / CAM (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Differentiate between CAD and CAM.
- (b) Differentiate between B-rep and C-rep.
- (c) Compare open loop and closed loop controls in NC machines.
- (d) Write short notes on rotary encoders.
- (e) Discuss the requirement of testing and calibration of NC machines.
- (f) Discuss the static and dynamic errors in NC machines.
- (g) With neat sketches discuss the elements of FMS system.
- (h) List any three programming language used in robots.



PART B

(4 × 15 = 60)

- II. (a) Discuss the various steps involved in FEM. (8)
- (b) Elaborate the levels of automation. (7)
- OR**
- III. (a) Discuss the significance of line balancing. (5)
- (b) Explain the algorithm of line balancing using Largest Candidate Rule. (10)
- OR**
- IV. (a) Write short notes on absolute and incremental co-ordinate systems. (7)
- (b) With an example illustrate the use of preparatory codes G02 and G03. (8)
- OR**
- V. Discuss the any five geometry statements and any five motion statements used in APT programming. (15)
- OR**
- VI. (a) With neat sketches explain about ATC used in CNC machines. (8)
- (b) With neat sketches explain about APC used in CNC machines. (7)
- OR**
- VII. Discuss the unique constructional features of NC machines. (15)
- OR**
- VIII. (a) Write short notes on lead through programming. (7)
- (b) Write short notes on vision sensors used in robots. (8)
- OR**
- IX. (a) With neat sketches explain the SCARA configuration of robots. (8)
- (b) Discuss the resolution, accuracy and repeatability of robots. (7)